

[PS 96] - Security Proofs for Signature Schemes
Pointcheval and Stern. Eurocrypt 1996

[BN 06] - Multi-Signatures in Plain Public Key Model and
a General Forging Lemma. Ballare and Neven. ACM 2006

[BCCGP 16] - Efficient Zero-knowledge Arguments for arithmetic
circuits in the Discrete Logarithm setting.
Boyle, Couteiro, Chaidos, Groth and Petit. Eurocrypt 2016.

[BBBPMW 18] - "Bulletproofs : Short proofs for confidential
transactions and More", Bünz, Boyle, Boneh,
Poelstra, Wuille and Maxwell, IEEE 2018

Why should I care about this topic?
usually/sometimes

→ We wish to prove security by stating something like this:

"If $\exists$ Adversary that breaks my protocol. Then I can break this believed-to-be hard math problem."
Ex: dlog, factoring, DDH etc...

Not always straight forward

↑
Forking lemma - a tool to achieve above

Intuitively: if A can break my protocol once he/she can do it twice. And I can break the believed-to-be-hard math problem if I receive 2 forgeries of the protocol.

↑
with same input but different response from the Random Oracle

↑
because it is only possible with one particular response from the R.O.
then it is so unlikely to get this precise response that I say it is negligible

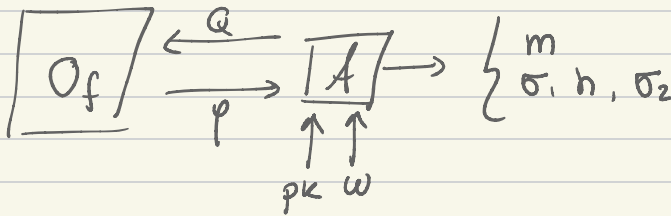
Forking lemma

Initially introduced by [PS96]

$$\text{Sign}(m, sk) \rightarrow (\sigma_1, h, \sigma_2)$$

- $h = \mathcal{H}(\sigma_1, m)$
- σ_2 depends only on σ_1, m, h
- Sometimes σ_1 & h can be omitted

No message attach



lemma "The forking lemma" [PS96] [PS96]

Let A be a probabilistic PPT Turing machine given only the public data as input. If A can find, with non-negligible probability a valid signature $(m, \sigma_1, h, \sigma_2)$, then with non-negligible probability a replay of this machine with the same random tape and a different oracle outputs two valid signatures $(m, \sigma_1, h, \sigma_2)$ & $(m, \sigma_1, h', \sigma_2')$ such that $h \neq h'$.

if only tree $\rightarrow$ one must be atleast $\epsilon/2$ likely

proof of lemma

Helping Lemma: Let $A: X \rightarrow Y$, such that $\Pr[A(x,y)] \geq \epsilon$, then there exists $Q \subset X$ s.t.:

- (i) $\Pr[x \in Q] \geq \epsilon/2$
- (ii) whenever $a \in Q$, $\Pr[A(a,y)] \geq \epsilon/2$

Assume no msg attacker
 A - PPT TM with random tape w .

During the attack this machine $q = \text{poly}(\lambda)$ questions to a RO O_f .
 Let $\begin{cases} Q_1, \dots, Q_q & \text{be the questions} \\ \varphi_1, \dots, \varphi_q & \text{be the answers} \end{cases}$

Random choice of the oracle $O_f \Rightarrow$ random choice of $\varphi_1, \dots, \varphi_q$

assumption

For a random choice of $w, \varphi_1, \dots, \varphi_q$, with non-negl prob A outputs a valid signature $(m, \sigma_1, h, \sigma_2)$

Remarks: $\circ$ It is easy to see that $\Pr[(m, \sigma_1) \notin \{Q_i\}_{i=1}^q] \leq \text{negl}$ due to $h = O_f(m, \sigma_1)$.

$\Pr[\text{success}] \geq \text{negl}$

$\Pr[Q_\beta] \geq \text{negl}$

For $\beta \in \{1, \dots, q\} \exists$ polynomial P s.t. probability of success over $w, \varphi_1, \dots, \varphi_q$ with $Q_\beta = (m, \sigma_1)$ is $\geq 1/P(\lambda)$

help lemma

Given such a $\beta : \xRightarrow{\text{lemma}} \exists$ a non-negl subset of Ω_β of good w 's.
 for such a good w the probability of success over $\varphi_1, \dots, \varphi_q$ with $Q_\beta = (m, \sigma_1) \geq 1/2 P(\lambda)$.

applying lemma again

Given such β and $w \xRightarrow{\text{lemma}} \exists$ a non-negl subset $R_{\beta, w}$ of 'good' $(\varphi_1, \dots, \varphi_{\beta-1})$'s. For such "good" answers the probability of success of the attacker over $\varphi_\beta, \dots, \varphi_q$ with $Q_\beta = (m, \sigma_1)$ is $\geq \frac{1}{2} \cdot \frac{1}{2P(\lambda)} = \frac{1}{4P(\lambda)}$.

Then with such β, w & $(\varphi_1, \dots, \varphi_{\beta-1})$ if we randomly choose $\rho_\beta, \dots, \rho_q$ & $(\rho'_\beta, \dots, \rho'_q)$, with a non negligible probability we obtain two valid signatures $(m, \sigma_1, h, \sigma_2) \neq (m, \sigma_1, h', \sigma'_2)$ s.t. $h \neq h'$
 $(K(\lambda) \gg \log \lambda)$

Finally with random choice of $\begin{cases} \beta, w, \varphi_1, \dots, \varphi_{\beta-1}, \rho_\beta, \dots, \rho_q \\ \beta, w, \varphi_1, \dots, \varphi_{\beta-1}, \rho'_\beta, \dots, \rho'_q \end{cases}$ with non-negl prob 2 valid signatures.

Helping lemma: Let $A \subset X \times Y$, such that $\Pr[A(x, y)] \geq \epsilon$, then there exists $\Omega \subset X$ s.t.:

- (i) $\Pr[x \in \Omega] \geq \epsilon/2$
- (ii) whenever $a \in \Omega$, $\Pr[A(a, y)] \geq \epsilon/2$

proof of lemma:

We define B_α to be:

$$B_\alpha = \{ (x, y) \in X \times Y \mid \Pr_{y \in Y}[(x, y) \in A] \geq \epsilon - \alpha \}$$

- by letting $\alpha = \epsilon/2$ & $\Omega := \{ x \in X : \forall y \in Y : (x, y) \in B_\alpha \}$
clearly $\Pr[B_\alpha] = \Pr[\Omega]$.

(i) Argument by contradiction.

let $\bar{B}$ be the complement of B in $X \times Y$
& assume $\Pr[B_\alpha] < \alpha$

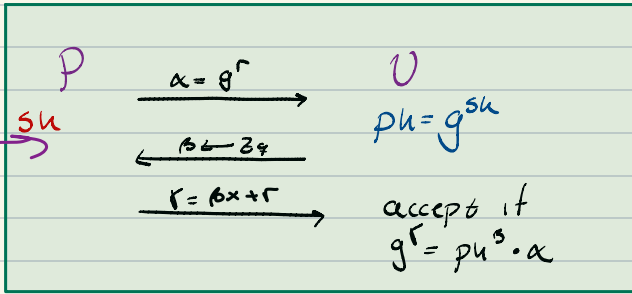
$$\begin{aligned} \epsilon &\leq \underbrace{\Pr[B] \cdot \Pr[A|B]}_{< \alpha} + \underbrace{\Pr[\bar{B}] \cdot \Pr[A|\bar{B}]}_{< \epsilon - \alpha} < \\ &= \Pr(A) \\ &< \alpha \cdot 1 + 1 \cdot \epsilon - \alpha = \epsilon \end{aligned}$$

$\Rightarrow$ contradiction by $\epsilon < \epsilon$!

$\rightarrow$ It also follows that $\Pr[\Omega] \geq \alpha$.
(ii) By definition of B_α .

Special Soundness + Schnorr ID Scheme

Rewinding P
bc here!



Remark α
the same
in both transcripts.

Given 2 ^{accepted} transcripts

$$\begin{cases} \tau = (\alpha, \beta, r) \\ \tau' = (\alpha, \beta', r') \end{cases}$$

we can extract x :

$$\text{if } \beta' \neq \beta \text{ then } x = \frac{r - r'}{\beta - \beta'}.$$

2 generalizations of the FL

[BN06] - generalizes the lemma to be more generic than a proof technique for Digital signatures.

Rewinds once

[BCCGPI6] - generalizes to consider more than one rewinding. Instead they consider an execution tree.

Rewinds as many times as needed

- unbounded runtime of extractor
but expected runtime is $T' \approx T$

↑
Runtime of
prover.

[BN06]

Generalized Forking Lemma

moving away from signatures

moving away from signatures.

This more general notion doesn't mention $\{-RC$ instead it focus on the output behaviour of $\{-$ signatures an algorithm when run twice on related inputs.

This makes it applicable to other contexts than signature schemes & separates the probabilistic analysis of the rewinding from the actual simulation in the security proof

Notation

x - public key
 $(h_1, \dots, h_q)$ - replies to queries to a RO.

queries to RO

hash function image space

LEMMA 1. [General Forking Lemma] Fix an integer $q \geq 1$ and a set H of size $h \geq 2$. Let A be a randomized algorithm that on input $x, h_1, \dots, h_q$ returns a pair, the first element of which is an integer in the range $0, \dots, q$ and the second element of which we refer to as a side output. Let IG be a randomized algorithm that we call the input generator. The accepting probability of A , denoted acc , is defined as the probability that $J \geq 1$ in the experiment

public key

$x \xleftarrow{\$} IG; h_1, \dots, h_q \xleftarrow{\$} H; (J, \sigma) \xleftarrow{\$} A(x, h_1, \dots, h_q)$.

The forking algorithm F_A associated to A is the randomized algorithm that takes input x proceeds as follows:

Algorithm $F_A(x)$

Pick coins ρ for A at random

$h_1, \dots, h_q \xleftarrow{\$} H$

$(I, \sigma) \leftarrow A(x, h_1, \dots, h_q(\rho))$

If $I = 0$ then return $(0, \varepsilon, \varepsilon)$

$h'_1, \dots, h'_q \xleftarrow{\$} H$

$(I', \sigma') \leftarrow A(x, h_1, \dots, h_{I-1}, h'_I, \dots, h'_q; \rho)$

If $(I = I' \text{ and } h_I \neq h'_I)$ then return $(1, \sigma, \sigma')$

Else return $(0, \varepsilon, \varepsilon)$.

Let

$$\text{frk} = \Pr \left[b = 1 : x \xleftarrow{\$} IG; (b, \sigma, \sigma') \xleftarrow{\$} F_A(x) \right].$$

Then

$$\text{frk} \geq \text{acc} \cdot \left(\frac{\text{acc}}{q} - \frac{1}{h} \right). \quad (1)$$

Alternatively,

$$\text{acc} \leq \frac{q}{h} + \sqrt{q \cdot \text{frk}}. \quad (2)$$

The transform can be used only if the protocol is public coin, that is when the random generated by the verifier is also known by the prover in the interactive version of the protocol.

[BCCG16]

Generalized Forking lemma

Tree of executions

The setting

- $(2\mu+1)$ -move public coin
- μ challenges $x_1, \dots, x_\mu$ in sequence
- $n_i > 1 \ \forall i \in \{1, \dots, \mu\}$
- $\prod n_i$ accepting transcripts with challenges in the following tree:
 - depth μ
 - $\prod n_i$ leaves
 - root is labelled with the statement
 - Each node at depth i has n_i children even labelled with a distinct value for the i^{th} challenge

Definition 7 (Statistical witness-extended emulation). (P, V) has statistical witness-extended emulation if for all deterministic polynomial time P^* there exists an expected polynomial time emulator $\mathcal{E}$ such that for all interactive adversaries A

$$\Pr \left[(u, s) \leftarrow \mathcal{A}(1^\lambda); tr \leftarrow \langle P^*(u, s), V(u) \rangle : \mathcal{A}(tr) = 1 \right] \\ \approx \Pr \left[(u, s) \leftarrow \mathcal{A}(1^\lambda); (tr, w) \leftarrow \mathcal{E}^{(P^*(u, s), V(u))}(u) : \right. \\ \left. \mathcal{A}(tr) = 1 \text{ and if } tr \text{ is accepting then } (u, w) \in R \right]$$

where the oracle called by $\mathcal{E}^{(P^*(u, s), V(u))}$ permits rewinding to a specific point and resuming with fresh randomness for the verifier from this point onwards.

Given an Adversary that produces an accepting argument with some probability, $\exists$ an emulator that produces a similar argument with some probability together with a witness w .

The emulator may rewind Prover/Verifier to any previous move.

Definition 8 (Public coin). An argument (P, V) is called public coin if the verifier chooses his messages uniformly at random and independently of the messages sent by the prover, i.e., the challenges correspond to the verifier's randomness ρ .

Above tree we referred to as an $(n_1, \dots, n_\mu)$ -tree of accepting transcripts.

generalisation of special-soundness for sigma protocols where $\mu=1$ $\Rightarrow \ell=2$



The Forking lemma [BCCGP16]

Let (P, V) be a $(2\mu+1)$ -move public coin interactive protocol. Let X be a witness extraction algorithm that always succeeds in extracting a witness from an $(n_1, \dots, n_\mu)$ -tree of accepting transcripts in PPT. Assume $\prod_{i=1}^{\mu} n_i \leq \text{poly}(\lambda)$. Then (P, V) has (statistical) witness-extended emulation.

Application of Generalized Forging Lemma

Bullet proofs [BBBPWM18]

Inner product argument:

An efficient proof system of the following relation:

$$\{g, h \in \mathbb{G}^n, P \in \mathbb{G}, c \in \mathbb{Z}_q : \underbrace{a, b \in \mathbb{Z}_q}_{\text{common input to both prover \& verifier}} : P = g^a h^b \wedge c = \langle a, b \rangle\}$$

input only known by Prover

The inner product protocol

- Input: $(g, h \in \mathbb{G}^n, u, P \in \mathbb{G}; a, b \in \mathbb{Z}_p^n)$ (7)
- $\mathcal{P}_{IP}$'s input: (g, h, u, P, a, b) (8)
- $\mathcal{V}_{IP}$'s input: (g, h, u, P) (9)
- Output: $\{\mathcal{V}_{IP}$ accepts or $\mathcal{V}_{IP}$ rejects
- (10)
- if $n = 1$: (11)
- $\mathcal{P}_{IP} \rightarrow \mathcal{V}_{IP} : a, b$ (12)
- $c = a \cdot b$ (13)
- $\mathcal{V}_{IP}$ checks if $P = g^a h^b u^c$: (14)
- if yes, $\mathcal{V}_{IP}$ accepts (15)
- otherwise, $\mathcal{V}_{IP}$ rejects (16)
- else: ($n > 1$) (17)
- $\mathcal{P}_{IP}$ computes: (18)
- $n' = \frac{n}{2}$ (19)
- $c_L = \langle a_{[n']}, b_{[n']} \rangle \in \mathbb{Z}_p$ (20)
- $c_R = \langle a_{[n']}, b_{[n']} \rangle \in \mathbb{Z}_p$ (21)
- $L = g^{a_{[n']}} h^{b_{[n']}} u^{c_L} \in \mathbb{G}$ (22)
- $R = g^{a_{[n']}} h^{b_{[n']}} u^{c_R} \in \mathbb{G}$ (23)
- $\mathcal{P}_{IP} \rightarrow \mathcal{V}_{IP} : L, R$ (24)
- $\mathcal{V}_{IP} : x \xleftarrow{\$} \mathbb{Z}_p^*$ (25)
- $\mathcal{V}_{IP} \rightarrow \mathcal{P}_{IP} : x$ (26)
- $\mathcal{P}_{IP}$ and $\mathcal{V}_{IP}$ compute: (27)
- $g' = g_{[n']}^{x^{-1}} \circ g_{[n']}^x \in \mathbb{G}^{n'}$ (28)
- $h' = h_{[n']}^{x^{-1}} \circ h_{[n']}^x \in \mathbb{G}^{n'}$ (29)
- $P' = L^{x^2} P R^{x^{-2}} \in \mathbb{G}$ (30)
- $\mathcal{P}_{IP}$ computes: (31)
- $a' = a_{[n']} \cdot x + a_{[n']} \cdot x^{-1} \in \mathbb{Z}_p^{n'}$ (32)
- $b' = b_{[n']} \cdot x^{-1} + b_{[n']} \cdot x \in \mathbb{Z}_p^{n'}$ (33)
- recursively run Protocol 2 on input (34)
- $(g', h', u, P'; a', b')$ (35)

Repeats this part 3 times

Sketch of the security proof.

only the parts relevant for Forging Lemma

An recursive argument that in each round we either extract a witness or a discrete logarithm relation.

$n = |g| - 1$: prover reveals the witness.

$n > 1$: Extractor runs prover to get L, R , then using 3 different challenges (x_1, x_2, x_3) , the extractor obtains: (pink part in)

$$a^{(1)}, b^{(1)}, a^{(2)}, b^{(2)}, a^{(3)}, b^{(3)} \text{ s.t.}$$

$$L^{x_1^2} P R^{x_1^{-2}} = g^{a^{(1)}} h^{b^{(1)}} u^{\langle a^{(1)}, b^{(1)} \rangle} \quad \forall i=1,2,3$$

From these 3 equalities the extractor can extract either a witness or a non-trivial discrete logarithm between g, h & u .